Supplementary material for: The Boussinesq–compressible gap in steady two-dimensional buoyancy-driven enclosure convection

Abstract

This supplement collects the tables that support the main text but are not needed in the flow of the argument: the asymptotic predictions and their numerical confirmation, the two-parameter fits, the exact single-ε transfer matrix, the learning curve, the public-data splits, the prefactor–velocity law, and the mapping from scripts to reported results. Table and section numbers in cross-references refer to the main text.

The runs behind every number in the main text are: 600 main solutions (`run_batch.sh`), 100 paired (`run_paired.sh`), 27 at $\varepsilon \to 0$ (`run_eps0.sh`), 104 at $Pr = 4.38$ (`run_water.sh`); `run_cpu_queue.sh` adds the 2:1 geometry and the tight-tolerance reruns.

Table S1: The five testable consequences of the expansion and their measured values.

prediction	measurement
$R \to 0$ as $\varepsilon \to 0$	intercept $+0.0003 \pm 0.0047$ (u_x); 0.06σ from zero
$R \propto \varepsilon$	$m = 1.023 \pm 0.018$ / 0.995 ± 0.019 / 1.009 ± 0.015 ($u_x/u_y/p$), window-averaged over the two decades; the residual is the ε^2 term (Sec. 5.1)
the shape of D does not rotate with ε	$\cos(D(\varepsilon), D(1.2)) = 0.95$–$1.00$, four channels $\times$ five values of ε
$\nabla \cdot \boldsymbol{u}_1 = \nabla \cdot (\theta_0 \boldsymbol{u}_0)$	$\mathrm{div}(Y)$ vs $\varepsilon \nabla \cdot (\theta \boldsymbol{u}_X)$: cosine $+0.994$ for $\varepsilon < 0.6$, $+0.965$ overall, amplitudes 2.26×10^{-2} vs 2.10×10^{-2}
the expansion converges fast	per configuration, ε^1 alone carries 86% of the correction energy, $\varepsilon^1 + \varepsilon^2$ carries 99.0%, ε^3 carries 99.9%

Table S2: Two-parameter refinement, $R = A\varepsilon^m Ra^q$. The Ra trend for u_y and θ reproduces in the 2:1 geometry.

channel	unit cavity ($n = 600$) m / q / R^2	aspect 2:1 ($n = 104$) m / q / R^2
u_x	1.081 / $+0.003 \pm 0.006$ / 0.74	1.071 / -0.143 ± 0.019 / 0.71
u_y	1.016 / -0.083 ± 0.005 / 0.77	1.146 / -0.077 ± 0.019 / 0.69
θ	1.120 / $+0.514 \pm 0.015$ / 0.72	0.989 / $+0.447 \pm 0.045$ / 0.55
p	1.127 / -0.029 ± 0.005 / 0.83	1.057 / -0.154 ± 0.017 / 0.74

Table S3: Exact transfer error $R(u_x)$ when a field computed at one ε (rows) is applied at another (columns).

$R(u_x)$ taken from ↓ / applied at →	0.4	0.8	1.2	1.6	1.9
$\varepsilon = 0.4$	0	0.067	0.143	0.235	0.353
$\varepsilon = 0.8$	0.068	0	0.078	0.183	0.305
$\varepsilon = 1.2$	0.149	0.081	0	0.110	0.243
$\varepsilon = 1.6$	0.265	0.199	0.112	0	0.140
$\varepsilon = 1.9$	0.402	0.328	0.251	0.142	0

Table S4: Learning curve on the fixed interpolation test set (200 samples), mean of three seeds. Entries are $u_x/u_y/\theta/p$. P3 keeps ε outside the network entirely.

n	B: ε as input	P3: predict $P(X)$, ε outside
0 (do nothing)	0.2203/0.2683/0.0381/0.2386	same
50	0.1831/0.1946/0.0269/0.1838	**0.1308/0.1364/0.0195/0.1501**
100	0.1280/0.1422/0.0199/0.1348	**0.0990/0.1022/0.0136/0.1150**
200	0.0974/0.1070/0.0147/0.0998	**0.0827/0.0866/0.0114**/0.1005
400	**0.0674/0.0714**/0.0097/**0.0672**	0.0735/0.0743/0.0093/0.0896

Table S5: Median relative correction on all five public splits, at $\varepsilon \equiv 1.2$.

split	n	u_x	u_y	θ	p
train (random walls)	4000	0.1797	0.2435	0.0228	0.1858
test_in_distribution	200	0.1822	0.2388	0.0220	0.1863
test_ood (harder wall dials)	200	0.1567	0.2157	0.0214	0.1671
train_legacy (classical cavity)	5000	0.2988	0.3362	0.0926	0.1989
test_legacy (classical cavity)	200	0.2984	0.3390	0.0836	0.1983

Table S6: The θ-channel prefactor is the flow strength: $R_\theta/\varepsilon = c\hat{u}^a$ on six data sets from two independent generators.

data set	n	$R_\theta/\varepsilon = c\hat{u}^a$	R^2
ours, random walls, $Pr = 0.71$	600	$a = 1.864 \pm 0.026$, $c = 0.544$	0.90
public train, random walls	4000	$a = 1.902 \pm 0.010$, $c = 0.602$	0.91
public test_in_distribution	200	$a = 1.977 \pm 0.041$, $c = 0.691$	0.92
public test_ood	200	$a = 1.931 \pm 0.041$, $c = 0.561$	0.92
public train_legacy, classical cavity	5000	$a = 2.360 \pm 0.002$, $c = 2.063$	**0.995**
public test_legacy, classical cavity	200	$a = 2.368 \pm 0.011$, $c = 2.095$	**0.996**

Table S7: Scripts and the results they produce.

script	what it produces
`exp_eps_law.py`	the 627-sample law (Sec. 5.1)
`exp_grid128.py`	the 64^2 vs 128^2 mesh check at $Ra = 10^4$ and $Ra = 10^6$ (Sec. 5.1)
`repack_p_fixed.py`	the corrected pressure channel (Sec. 5.2)
`exp_paired_final.py`	the ε-blind floor and shape invariance (Sec. 5.3)
`exp_modes.py`	the ε^k mode expansion (Sec. 3)
`exp_chi.py`	the χ explanation for θ (Sec. 5.1)
`harness.py`, `exp_ml_final.py`	the unified surrogate ablation: A/B/P1/P2/P3, three seeds (Secs. 5.4–5.6)
`fno_lib.py`, `exp_fno_final.py`	the FNO baseline under the same protocol (Sec. 5.5)
`exp_lc_final.py`	learning curves (Sec. 5.6)
`exp_pod.py`	the POD-ROM / linear-surrogate baselines (Sec. 5.4)
`exp_direct_linear.py`, `exp_direct_batch.py`	the direct solution of the linearised $O(\varepsilon)$ system on all 20 configurations (Sec. 5.4)
`exp_direct_eps0.py`	the same one-Newton-step solve on the 27 low-contrast samples (Sec. 5.4)
`exp_heldout_loweps.py`, `run_loweps.sh`	the 38 held-out solutions at $\varepsilon = 0.1, 0.2$ and the extrapolated-P check (Sec. 5.4)
`exp_direct_low.py`, `exp_fixed_flow_eps.py`	one-Newton-step solves at $\varepsilon = 0.1, 0.2$ on the paired flows and the fixed-flow ε-drift (Sec. 5.4)
`exp_cluster_boot.py`	family-cluster bootstrap of the exponent (Sec. 5.1)
`exp_p2_fixed.py`, `exp_prior_ablation.py`, `exp_fno.py`, `exp_learning_curve.py`	first-generation runs of the same experiments (superseded by the three scripts above; their JSON outputs were deleted as stale)
`exp_public_transfer.py`	external validation and transfer (Secs. 5.7–5.9)
`exp_aspect2.py`	the 2:1 geometry (Sec. 5.10)
`exp_tight.py`, `verify_noise.py`, `exp_final_numbers.py`	noise control and timing (Secs. 5.11–5.12)
`exp_water.py`	the strictly paired $Pr = 4.38$ fluid study (Sec. 5.13)
`exp_perturbation.py`	the first-order perturbation baseline on the ε scan (Sec. 5.4)
`exp_linear_response.py`	the linear-response and secant baselines from the paired scan (Sec. 5.4)
`exp_second_order.py`	the two-term $P + \varepsilon Q$ fit from the paired scan (Sec. 5.4)
`exp_nu_clean_summary.py`	the isothermal-sidewall Nu statistics and the denominator audit (Sec. 5.8)
`exp_nu_grid.py`, `exp_nu_grid_ra1e6.py`, `exp_nu_grid_ra1e5.py`	the 64^2 versus 128^2 wall-flux check (Sec. 5.1)
`exp_public_nu.py`, `exp_flux_loss.py`, `exp_flux_stencil.py`	wall heat transfer of the public-data surrogate, with and without the wall-flux training penalty, and the cross-stencil check (Sec. 5.8)
`exp_film.py`	the FiLM-conditioned baseline (Sec. 5.5)
`exp_3d_law.py`, `exp_3d_exp.py`, `exp_2d_ra_window.py`, `merge_3d.py`, `exp_3d_summary.py`, `make_fig9.py`, `exp_3d_rom_eps.py`	the twenty-four-sample three-dimensional check at two Rayleigh numbers and the POD-ROM baseline (Sec. 5.14)
`verify_conventions.py`	the 38 400-value convention audit against the published `relL2` columns (Sec. 5.7)
`exp_public_splits.py`, `exp_legacy5000.py`, `exp_uhat_law.py`	the five public splits and the $\hat{u}$ law (Sec. 5.7)
`exp_transfer_uhat.py`	the $\hat{u}$ transfer criterion on the public splits (Sec. 5.9)
`audit_v2.py`	robustness checks: m vs fitting window, κ, pressure-pair conventions, $\hat{u}$ medians, the 2:1 σ distances
`make_figs.py`, `make_figs2.py`	every figure, from the JSON results
`make_tables.py`	regenerates every table from the JSON results